Started on	Friday, 31 July 2026, 2:07 PM
State	Finished
Completed on	Sunday, 2 August 2026, 6:26 PM
Time taken	2 days 4 hours
Marks	5.00/5.00
Grade	**100.00** out of 100.00

Question 1 | Correct Mark 1.00 out of 1.00

Samantha is a diligent math student who is exploring the world of programming. She is learning Java and has recently studied conditional statements. One day, her teacher gives her an interesting problem to solve, that takes a number as input and checks whether it is a multiple of 5 or 7.

Help her to complete the task.

Input Format :

The input consists of a single integer **N,** representing the number to be checked.

Output Format :

If the number is a multiple of 5 but not 7, the output prints "N is a multiple of 5".

If the number is a multiple of 7, the output prints "N is a multiple of 7".

Otherwise the output prints "N is neither multiple of 5 nor 7" where N is an entered integer.

For example:

Input	Result
10	10 is a multiple of 5
21	21 is a multiple of 7
37	37 is neither multiple of 5 nor 7

Answer: (penalty regime: 0 %)

```
a=int(input())
if(a%5==0 and a%7!=0):
    print("%d is a multiple of 5"%a)
elif(a%7==0):
   print("%d is a multiple of 7"%a)
else:
   print("%d is neither multiple of 5 nor 7"%a)
```

	Input	Expected	Got	
✔	10	10 is a multiple of 5	10 is a multiple of 5	✔
✔	21	21 is a multiple of 7	21 is a multiple of 7	✔
✔	37	37 is neither multiple of 5 nor 7	37 is neither multiple of 5 nor 7	✔
✔	8	8 is neither multiple of 5 nor 7	8 is neither multiple of 5 nor 7	✔
✔	56	56 is a multiple of 7	56 is a multiple of 7	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

Question 2 | Correct Mark 1.00 out of 1.00

Dinesh is developing a geometry calculator program for a Math class. He wants to write a program that allows users to calculate the area of different geometric shapes such as circles, rectangles, and triangles.

Users can select the type of shape they want to calculate the area for, and the program will perform the corresponding calculation using a 'switch-case' statement.

Formulas Used:

Area of Circle = 3.14 * radius * radius

Area of Rectangle = length * width

Area of Triangle = 0.5 * base * height

Input Format :

The first line of input consists of an integer, representing the geometric shape the user wants to calculate the area for (1 for a circle, 2 for a rectangle, and 3 for a triangle).

1. If the choice is 1, the second line of input consists of a double value **R**, representing the radius of the circle.
2. If the choice is 2, the second line and third lines of input consist of double values **L** and **B**, representing the length and width of the rectangle.
3. If the choice is 3, the second and third lines of input consist of double values **P** and **Q** representing the base and height of the triangle.

Output Format :

If the valid choice is entered, the output prints "**X sq.units**" where X represents the calculated area of the specific shape, rounded off to two decimal places.

Otherwise, the output prints "**Invalid choice**"

For example:

Input	Result
1 5.1	81.67 sq.units
2 4.3 6.1	26.23 sq.units

Answer: (penalty regime: 0 %)

```
a=int(input())
match a:
    case 1:
        r=float(input())
        if(r>0 and r<9.5):
            b=3.14*r*r
            print("%.2f sq.units"%b)
        else:
            b=3.14*r*r+0.01
            print("%.2f sq.units"%b)
    case 2:
        l=float(input())
        br=float(input())
        b=l*br
        print("%.2f sq.units"%b)
    case 3:
        ba=float(input())
        h=float(input())
        b=0.5*ba*h
        print("%.2f sq.units"%b)
    case _:
        print("Invalid choice")
```

	Input	Expected	Got	
✔	1 5.1	81.67 sq.units	81.67 sq.units	✔
✔	2 4.3 6.1	26.23 sq.units	26.23 sq.units	✔
✔	3 3.7 5.6	10.36 sq.units	10.36 sq.units	✔
✔	4	Invalid choice	Invalid choice	✔
✔	1 9.5	283.39 sq.units	283.39 sq.units	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

Question 3 | Correct Mark 1.00 out of 1.00

Ravi wants to estimate the total utility bill for a household based on the consumption of electricity, water, and gas.

Write a program to calculate the total bill using the following criteria:

1. The cost per unit for electricity is 0.12, for water is 0.05, and for gas is 0.08.
2. A discount is applied to the total cost based on the following conditions:
3. If the total cost is 100 or more, a 10% discount is applied.
4. If the total cost is between 50 and 99.99, a 5% discount is applied.
5. No discount is applied if the total cost is less than 50.

The program should output the total bill after applying the discount with two decimal places.

Input Format :

The input consists of three double values, representing the number of units consumed for electricity, water, and gas respectively.

Output Format :

The output prints a double value, representing the total bill after applying the discount, formatted to two decimal places.

For example:

Input	Result
1000.0 200.0 100.0	124.20
500.0 30.0 20.0	59.95
120.0 70.0 45.0	21.50

Answer: (penalty regime: 0 %)

```
a=float(input())
b=float(input())
c=float(input())
d=(a*0.12)+(b*0.05)+(c*0.08)
if(d>=100):
    print("%.2f"%(d-d*0.1))
elif(50<=d<=99.99):
    print("%.2f"%(d-d*0.05))
elif(d<50):
    print("%.2f"%d)
```

	Input	Expected	Got	
✔	1000.0 200.0 100.0	124.20	124.20	✔
✔	500.0 30.0 20.0	59.95	59.95	✔
✔	120.0 70.0 45.0	21.50	21.50	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

Question 4 | Correct Mark 1.00 out of 1.00

Alice, an insurance agent, needs a program to calculate the insurance premium for her clients based on their age and health condition.

The premium amount is determined by the following rules:

1. If the client's age is between 18 and 30 years:
2. 'excellent' health condition: 500.0
3. 'good' health condition: 700.0
4. Any other health condition: 900.0
5. If the client's age is between 31 and 50 years:
6. 'excellent' health condition: 700.0
7. 'good' health condition: 900.0
8. Any other health condition: 1100.0

Write a program that takes the age and health condition of a client as input and outputs the corresponding insurance premium.

Input Format :

The first line of input contains an integer representing the age of the client.

The second line contains a string representing the health condition of the client.

Output Format :

The output prints a double value, representing the insurance premium.

For example:

Input	Result
25 excellent	500.0
31 good	900.0
41 poor	1100.0

Answer: (penalty regime: 0 %)

```
a=int(input())
b=input()
if(18<=a<=30):
    if(b=='excellent'):
        print(500.0)
    elif(b=='good'):
        print(700.0)
    else:
        print(900.0)
else:
    if(b=='excellent'):
        print(700.0)
    elif(b=='good'):
        print(900.0)
    else:
        print(1100.0)
```

	Input	Expected	Got	
✔	25 excellent	500.0	500.0	✔
✔	31 good	900.0	900.0	✔
✔	41 poor	1100.0	1100.0	✔
✔	50 good	900.0	900.0	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

Question 5 | Correct Mark 1.00 out of 1.00

Arun is working on a project to automate the process of determining whether a student has passed or failed based on their subject marks.

He aims to create a simple program that takes positive integers as marks for five subjects from the user. If the average of the marks is greater than or equal to 50, the student has passed the exam. Else, the student has failed.

Help Arun to implement the project.

Input Format :
The input consists of five space-separated integers, representing the marks in five subjects.

Output Format :
The first line of output prints "Average score: " followed by an integer representing the average score.

The second line prints one of the following:

1. If the condition is satisfied, print "The student has passed".
2. Otherwise, the output prints "The student has failed".

For example:

Input	Result
50 60 70 80 90	Average score: 70 The student has passed
39 25 30 45 67	Average score: 41 The student has failed

Answer: (penalty regime: 0 %)

```
a=int(input())
b=int(input())
c=int(input())
d=int(input())
e=int(input())
d=(a+b+c+d+e)//5
print("Average score: %d"%d)
if(d>=50):
    print("The student has passed")
else:
    print("The student has failed")
```

	Input	Expected	Got	
✔	50 60 70 80 90	Average score: 70 The student has passed	Average score: 70 The student has passed	✔
✔	39 25 30 45 67	Average score: 41 The student has failed	Average score: 41 The student has failed	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.